Keenan Ganz

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OBJECTIVE: To promote responsible natural resource management via software and field research.

EDUCATION

RENSSELAER POLYTECHNIC INSTITUTE (4.0 / 4.0 GPA CUMULATIVE) **EXPECTED GRADUATION 2022**

ENVIRONMENTAL SCIENCE & BIOCHEMISTRY B.S., GEOINFORMATICS CONCENTRATION

Relevant Coursework: Geology II: Earth's Surface, Earth's Climate, Computer Science I (Python), Data Structures (C++), Biostatistics (R)

WORK EXPERIENCE

ENVIRONMENTAL SCIENCE ASSOCIATES **MAY 2020 – AUG 2020**

GEOSPATIAL SERVICES INTERN

Position supporting ESA's geospatial analytics and firm-wide data science initiative. Projects focused on detection of vegetation, wetlands, and open water from remotely-sensed imagery (NAIP and Sentinel-2) and ground-truth observations. Tools included ArcGIS, R, and Google Earth Engine.

SNOHOMISH COUNTY PUBLIC WORKS **MAY 2019 – AUG 2019**

ENVIRONMENTAL SERVICES INTERN

Seasonal position supporting environmental code enforcement and field monitoring of mitigation areas (plots replanted post-construction) in suburban Washington. Tasks included a mixture of surveying, data analysis, and independent GIS projects.

RENSSELAER ASTROBIOLOGY RESEARCH AND EDUCATION CENTER **JAN 2019 - PRESENT**

RESEARCH ASSISTANT

Employed as both research assistant and work-study student. Work focuses on instrumentation for spontaneous polymerization of RNA in an abiotic environment (i.e. the RNA World Hypothesis). Developed method for quantifying polymer length that distinguished between cyclic aggregates and true straight-chain polymers using electrospray ionization mass spectrometry (ESI-MS). Identified three intermediate polymer species undetected using conventional techniques (see Awards section).

UNIVERSITY OF WASHINGTON SCHOOL OF OCEANOGRAPHY **MAY 2018 – AUG 2018**

VOLUNTEER DEVELOPER

Development and testing of yodapy, a Python package to broaden access to Ocean Observatories Initiative, a database of hundreds of real-time oceanographic sensors. Package has over 500 downloads to date; use cases for web-based and local analysis in Jupyter notebooks were presented at OceanHackWeek 2018 to demonstrate package to new users.

SERVICE ORGANIZATIONS

CODING&&COMMUNITY **MAY 2019 – PRESENT**

Student-driven computer science outreach and education organization based at Rensselaer Polytechnic Institute and dedicated to closing disparities in education related to race, gender, and socioeconomic status in upstate New York. Teaching programs through RPI's Science and Technology Entry Program, Troy High School, Troy Middle School, and Rise High. Organization records 300+ contact hours of instruction per academic year.

Leadership Positions:

- President, May 2020 – Present
  - Currently shifting club operations to a remote learning environment. Training instructor corps on how to teach computer science effectively without in-person instruction. Migrating curriculum base to platforms accessible on-demand.
- Director of External Operations, May 2019 - May 2020
  - Planned Launchpad, annual hackathon for middle to high school students. Standardized curriculum development process and developed new course on web development.
- STEP Instructor, September 2019 - Present
  - Teaches introductory computer science concepts to middle and early high school students. Curriculum developer for Python and web development courses.

AWARDS

RPI UNDERGRADUATE RESEARCH SYMPOSIUM, 2020

1ST PLACE POSTER (SCIENCES)

Award given for research project entitled "Instrumentation for Abiotic RNA Oligomerization Analysis". Compiled research results into static 6-slide PowerPoint to be presented remotely.

NATIONAL MERIT SCHOLAR

Award given to incoming college students following an essay application and scoring in the top 1% in their state on the SAT.